Assignment 11

Module -13 Windows Networking Services

1. Discuss the role of Windows Firewall in Windows Server and how to configure it.

➢ windows firewall is a security feature in Windows Server that protects the server from unauthorized access and harmful network traffic.

❖ **It monitors:**

- Incoming traffic
- Outgoing traffic
- Allowed and blocked connections

❖ **Why is it Important?**

- It prevents hackers from accessing the server.
- It blocks suspicious traffic.
- It allows only trusted applications and services.
- It improves overall network security.

➢ Without a firewall, any device on the network may try to access the server, which can lead to security attacks.

❖ **How to Configure Windows Firewall**

Step 1: Open Firewall

- Open Server Manager
- Go to Tools
- Select Windows Defender Firewall with Advanced Security

Step 2: Configure Rules

You can create:

- Inbound Rules → Control incoming traffic
- Outbound Rules → Control outgoing traffic

Step 3: Create a New Rule

- Click Inbound Rules
- Select New Rule
- Choose:
  - Port
  - Program
  - Service

➢ So the firewall protects the server but still allows required services.

2. What is Network Address Translation (NAT) in Windows Server, and how do you configure it?

   - Network Address translation (NAT) is a networking process that converts private IP addresses into public IP addresses.
   - It allows many computers in a private network to share one internet connection.

   - **Why NAT is Used**

   - NAT is used because:
     - Private IP addresses cannot directly access the internet.
     - It saves public IP addresses.
     - It hides internal network addresses for security.

   - **How to Configure NAT in Windows Server**

Step 1: Install Remote Access Role

- Open Server Manager
- Click Add Roles and Features
- Select:
  - Remote Access
  - Routing

Step 2: Open Routing and Remote Access

  - Go to Tools
  - Open Routing and Remote Access

Step 3: Enable NAT
- Right-click server name
- Select:
  - Configure and Enable Routing and Remote Access

Step 4: Select NAT
- Choose:
  - Network Address Translation (NAT)

Step 5: Select Internet Interface
- Select the interface connected to the internet
- Enable:
  - Public Interface
  - Enable NAT

   - Now all internal devices can access the internet through one public IP.

3. Explain the concept of Dynamic Host Configuration Protocol (DHCP) and how to configure it in Windows Server 2016.

   - Dynamic Host Configuration Protocol (DHCP) automatically provides IP addresses to devices in a network.

     - **Without DHCP:**
       - Every computer must be configured manually.
     - **With DHCP:**
       - IP settings are assigned automatically.

     - Why DHCP is Important
       - DHCP reduces:
         - Manual work
         - IP conflicts
         - Configuration errors
       - It automatically provides:
         - IP address
         - Subnet mask
         - Default gateway
         - DNS server
     - **How to Configure DHCP in Windows Server 2016**

Step 1: **Install DHCP Role**

- Open Server Manager
- Select Add Roles and Features
- Choose:
  - DHCP Server

Step 2: **Complete Post-Installation**

- Click notification flag
- Select:
  - Complete DHCP Configuration

Step 3**: Create DHCP Scope**

- Open DHCP
- Expand IPv4

Step 4: **Configure Scope**

Enter:

- Scope name
- IP range
- Subnet mask
- Gateway
- DNS server

   - Now devices automatically receive IP addresses from the DHCP server.

4. Describe the process of configuring DNS (Domain Name System) in Windows Server.

- DNS (Domain Name System) converts domain names into IP addresses.
  - Example:
    - google.com → 142.x.x.x
- Humans remember names easily, while computers communicate using IP addresses.

  - **Why DNS is Important**
    - Users must remember IP addresses.
    - DNS makes internet and network access easier.

  - **How to Configure DNS in Windows Server**

Step 1: **Install DNS Role**
- Open Server Manager
- Select Add Roles and Features
- Choose:
  - DNS Server

Step 2: **Open DNS Manager**
- Go to Tools
- Open DNS

Step 3**: Create Forward Lookup Zone**
- Right-click:
  - Forward Lookup Zones
- Select:
  - New Zone

Step 4: **Configure Zone**
- Choose:
  - Primary Zone
- Enter domain name
  - Example: company.local

Step 5: **Add Host Record**
- Right-click zone
- Select:
  - New Host (A or AAAA)

Enter:
- Host name
- IP address

- Now users can access:
- server1.company.local
  instead of typing the IP address.

5. What is Server Manager, and how do you use it to manage servers in Windows Server?

- ➢ Server Manager is a management tool in Windows Server used to manage server roles, features, and services from one location.

❖ **Why Server Manager is Important**

➢ It helps administrators:

- Install roles and features
- Monitor server health
- Configure networking
- Manage multiple servers
- Restart services

❖ **Main Functions of Server Manager**

1.**Add Roles and Features**

Install:

- DHCP
- DNS
- Active Directory
- Web Server (IIS)

2. **Monitor Server Status**

View:

- Events
- Performance
- Alerts

3**. Manage Multiple Servers**

4. **Access Administrative Tools**

Direct access to:

- DHCP
- DNS
- Firewall
- Computer Management

❖ **How to Use Server Manager**

Step 1: Open Server Manager

Step 2:Select required section:

- Local Server
- All Servers
- Tools

Step 3:Perform management tasks.

6. Discuss the role of Remote Desktop Services (RDS) in Windows Server 2016 or 2019 and how to configure it. give me a ans and explain why it is possible and explain it

- ➢ Remote Desktop Services (RDS) allows users to remotely access another computer or server over a network.
- ❖ Users can:
  - Control the server remotely
  - Run applications
  - Manage files and services

❖ **Why RDS is Important**

- Administrators can manage servers remotely.
- Employees can work from different locations.
- Multiple users can access the same server.

It reduces the need for physical access to the server.

❖ **How to Configure RDS**

Step 1: Install RDS Role

- Open Server Manager
- Click:
  - Add Roles and Features
- Select:
  - Remote Desktop Services Installation

Step 2: Choose Deployment Type

- Quick Start

Step 3: Select Server Roles

Install:

- Session Host
- Licensing
- Connection Broker

Step 4: Enable Remote Desktop

- Go to:
  - This PC → Properties → Remote Settings
- Enable:
  - Allow Remote Connections

Step 5: Connect Remotely

Remote Desktop Connection

Enter:

- Server IP
  or
- Hostname
  - ➢ Users can remotely log in and manage the server from another compute